Does College Students' Alcohol Consumption Have a Negative Effect on Academics

This report used data from the 2022 MACHB survey of Truman students. The MACHB survey gathered demographics and asked students about their current health behaviors. I chose to take a more in depth look at the responses of alcohol consumption, to see if it had a negative effect on student's academics.

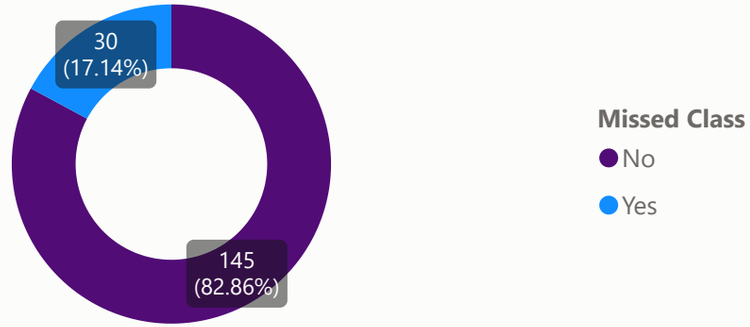
I looked at 2 outcome variables that were part of the same question on the survey. The questions from the survey asked students how often in the past 12 months have they performed poorly on a test or assignment due to alcohol and how many times they missed class due to alcohol. An initial glance at the data showed that around 90% of students reported 0 times for these questions, with small percentages making up the rest of the options. Because of this, I decided to group the responses into a binary variable for my analysis, tracking whether or not the student had experienced at least 1 academic consequence due to alcohol. Additionally, with all the data, I only looked at the responses where the student reported that they had consumed alcohol in the past 12 months, because these questions wouldn't apply to the people who didn't drink at all.

Total Students Who Consumed Alcohol in the Past 12 Months

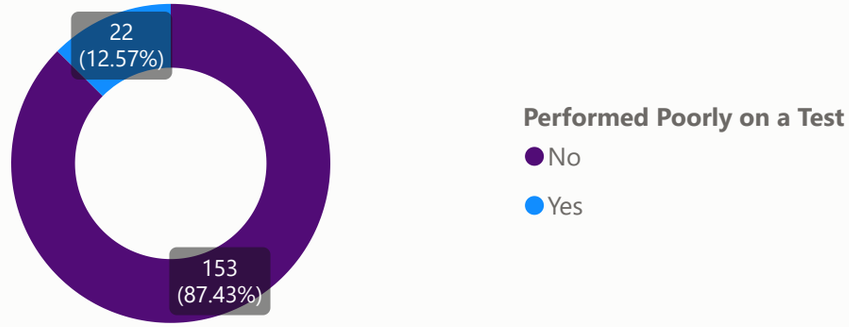
175

These pie charts show how many students missed class and performed poorly on a test due to alcohol. Only a small proportion suffered these consequences from their alcohol consumption, but it does effect some students, and is something that could be improved to help students succeed more in the classroom.

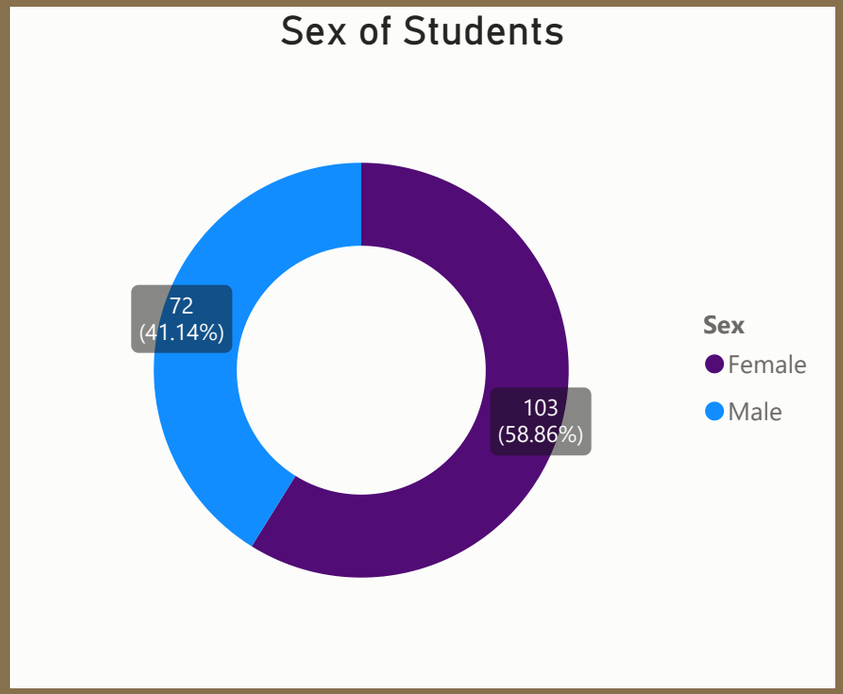
Have Students Missed Class as a Result of Alcohol in the Past 12 Months

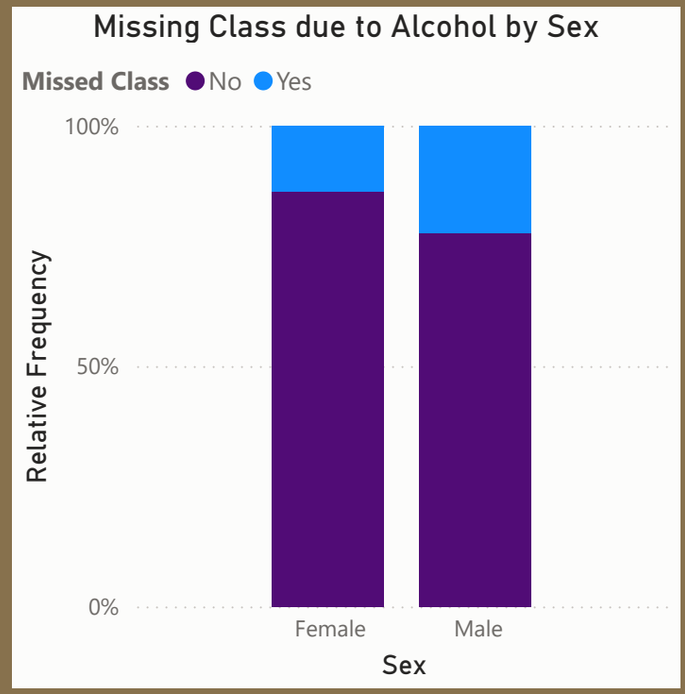


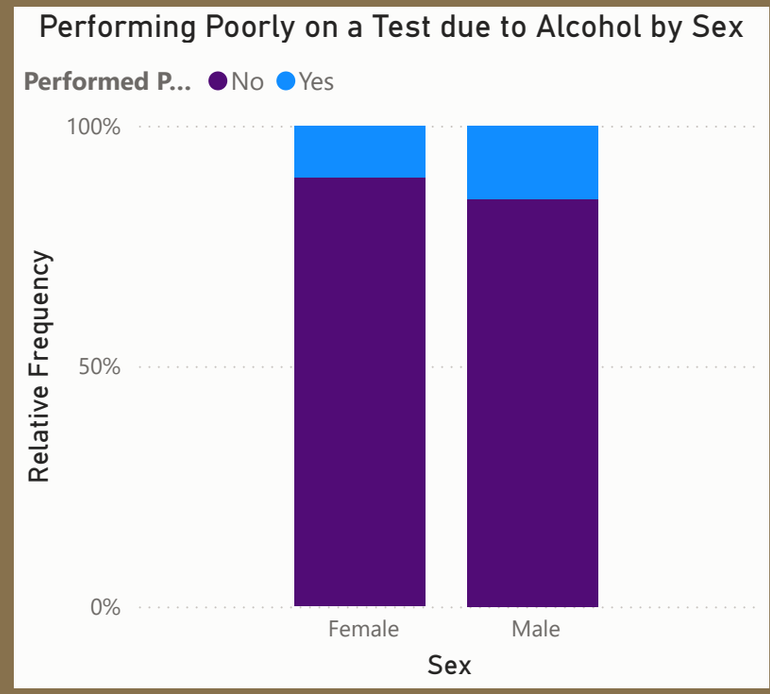
Have Students Performed Poorly on a Test as a Result of Alcohol in the Past 12 Months



The Effect of Sex on Negative Academic Consequences From Alcohol







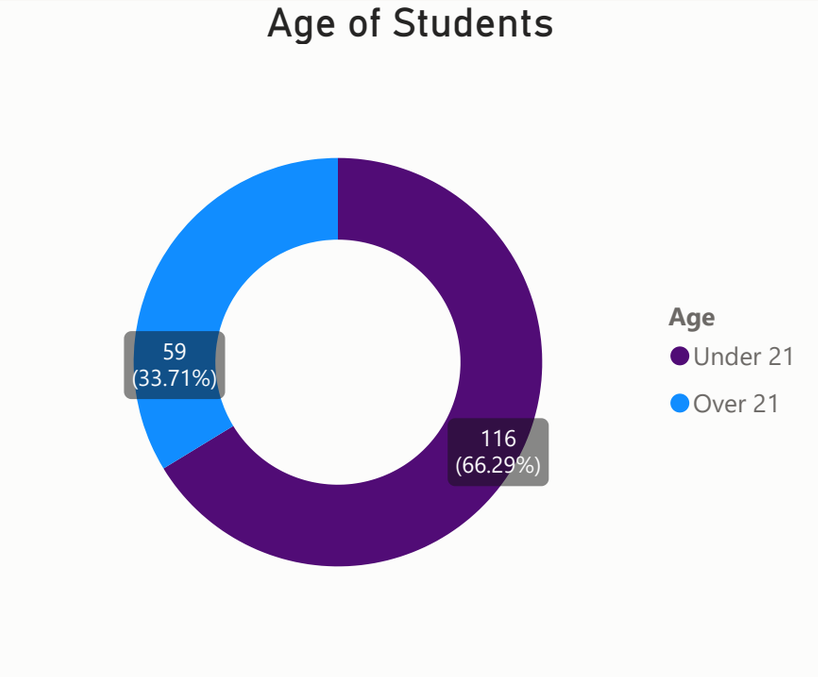
When looking at Sex, I removed the "Prefer not to Respond" rows because there were so few of them and only focused on Male/Female

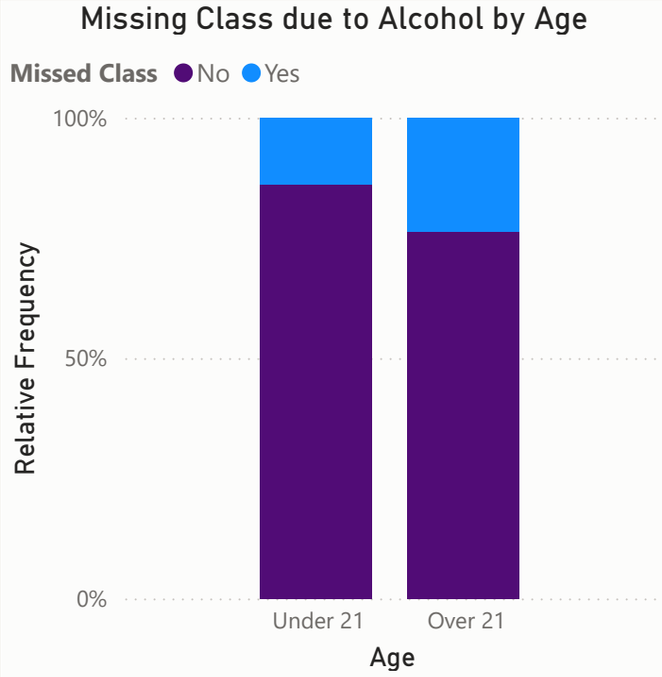
Of the students in this subset of the data, 58.86% were Female and 41.14% were Male

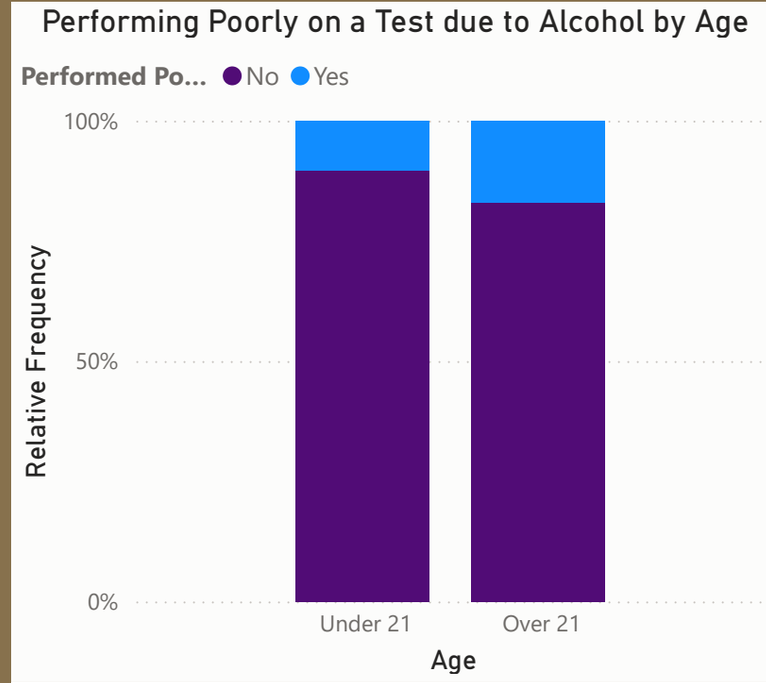
These graphs show that males more commonly suffer academic consequences due to alcohol than females, even if the difference is slight.

This agrees with what I would have expected, that males on average can be a little more irresponsible than females when dealing with alcohol and its effects.

The Effect of Age on Negative Academic Consequences From Alcohol







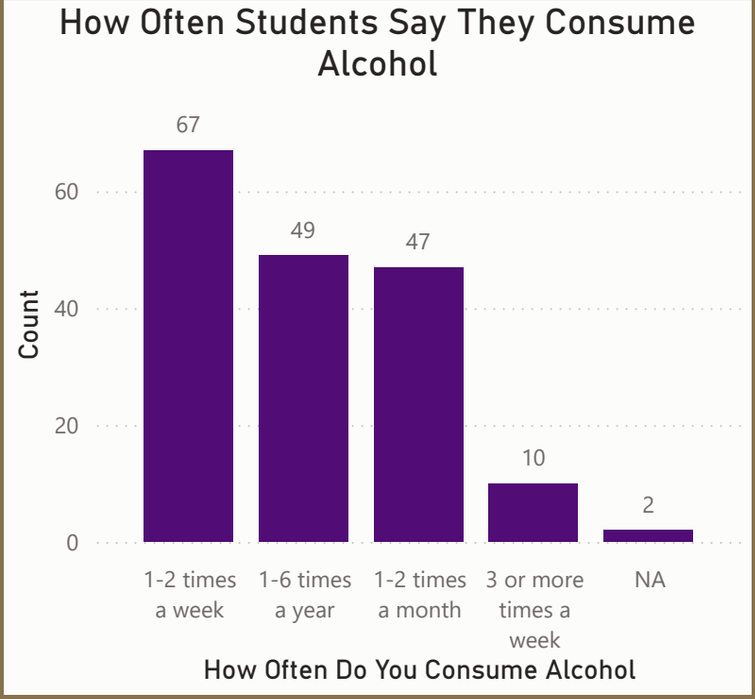
When looking at Age, I chose to only look at whether a student had turned 21 and was of legal drinking age or not.

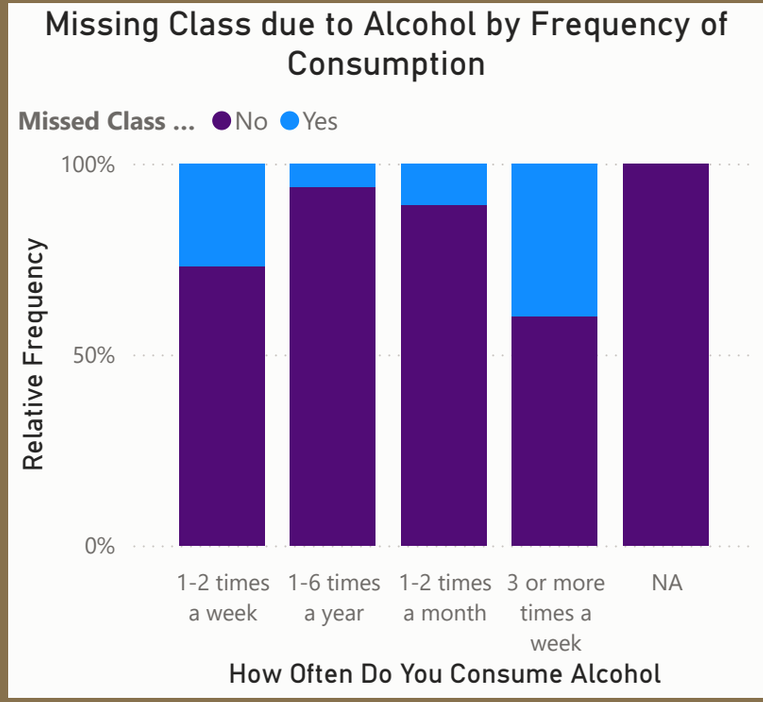
Of the students in this subset of the data, 66.29% were Under 21 and 33.71% were Over 21

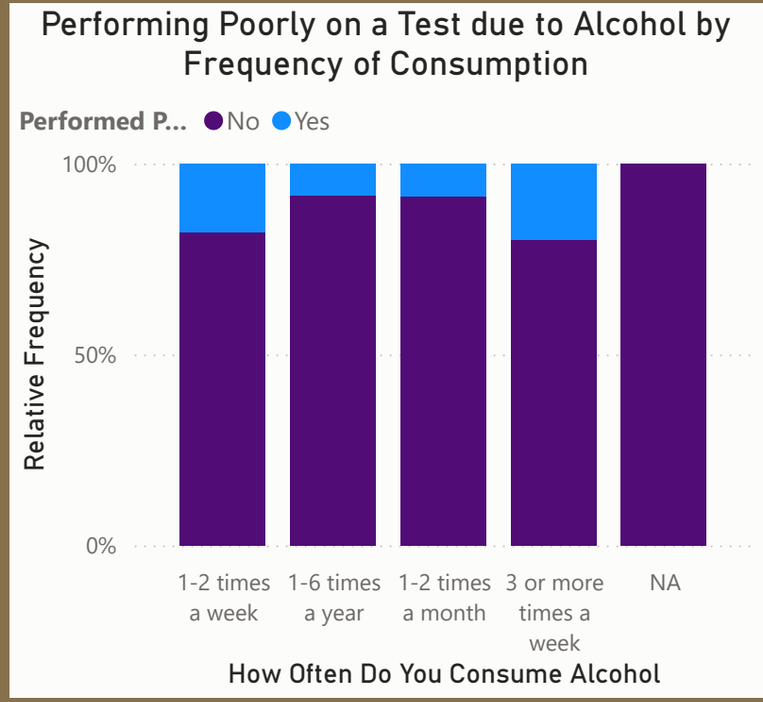
These graphs show that college students over 21 that drink alcohol on average are more likely to suffer academic consequences due to alcohol than those who are under 21.

This was surprising to me as I would have guessed students under 21 would be less responsible and more likely to not manage their drinking well. What I assume is the cause for the results are students over 21 have easier access to alcohol and its easier for it to get out of hand.

The Effect of Frequency of Consumption on Negative Academic Consequences From Alcohol





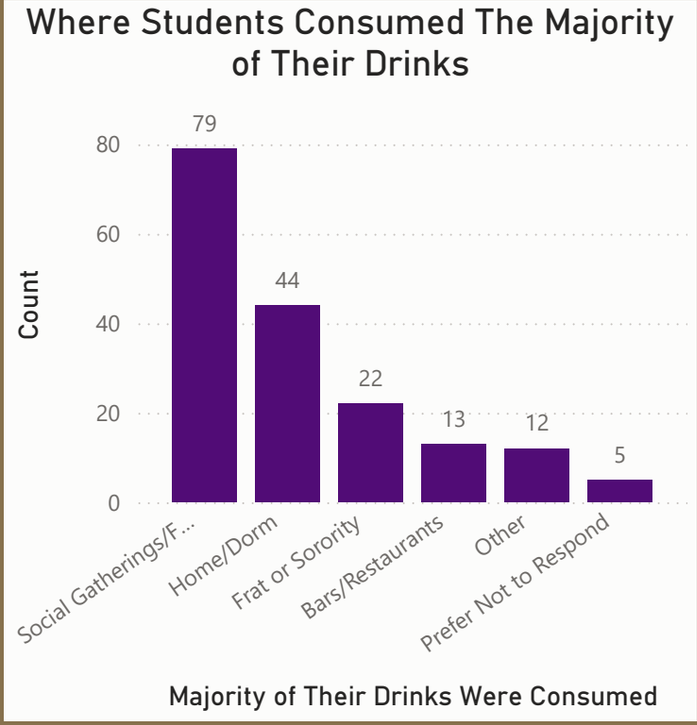


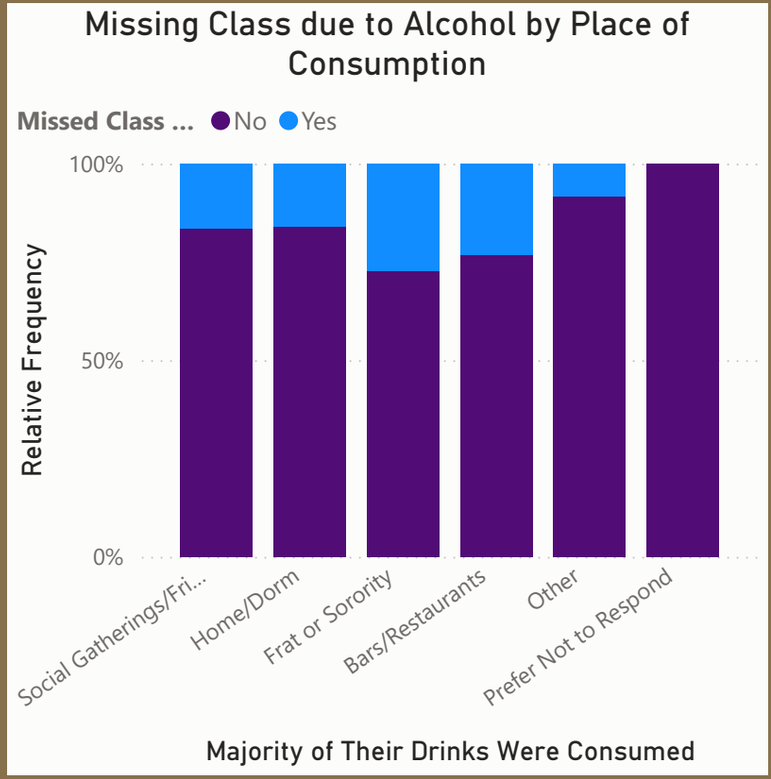
While the most common choice was 1-2 times per week, over half of the students that drank were occasional drinkers, only drinking 1-2 times a month or less.

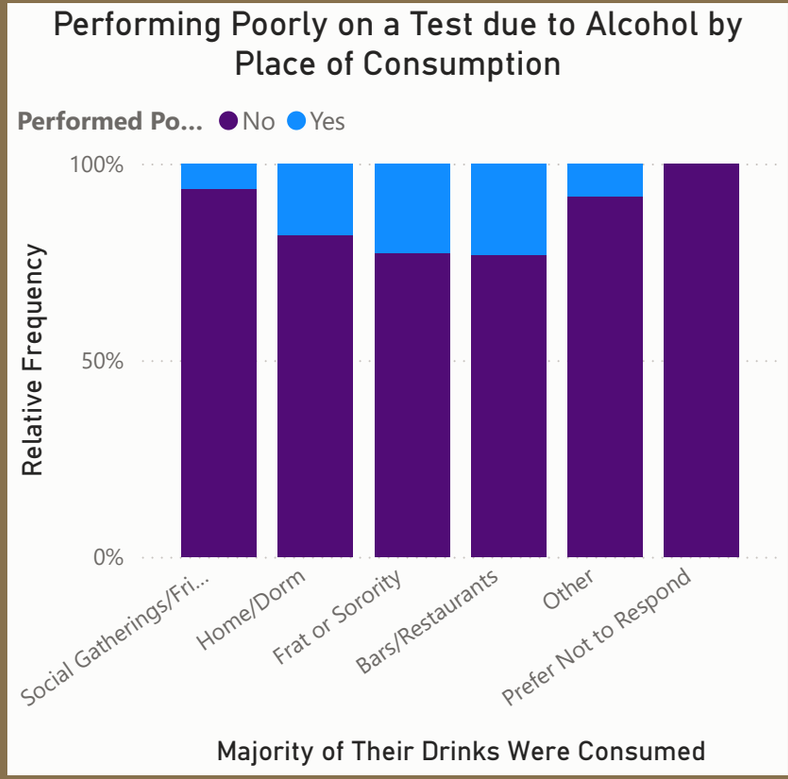
From these stacked bar charts, we can see that a greater proportion of frequent drinkers (1-2 times a week and 3 or more times a week) struggle with missing class and performing poorly on tests due to alcohol than less frequent drinkers (1-2 times a month and 1-6 times a year).

This makes sense, as students who drink more have more opportunities where alcohol can cause them negative academic consequences.

The Effect of the Place Where Students Drink the Most on Negative Academic Consequences From Alcohol







This graph shows most students consume a majority of their drinks at social gatherings or friends houses. Also common is drinking at home, and less common is drinking mostly at a frat or sorority, bar/restaurant, or other place.

All of these places have some students who drink there and suffer some academic consequences because of alcohol, but it seems that students who drink mostly at frats/sororities and bars/restaurants are more likely to suffer these academic consequences than students who drink at home and at friends house, but the differences are slight.